Response to Reviewers

GraphyloVar: Predicting the impact of non-coding variants using a multi-species sequence model

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Dongjoon Lim and Mathieu Blanchette

We thank the reviewers for their reading and feedback. We addressed the comments. New and revised text in the manuscript is indicated by blue highlighting in the revised PDF. Below we respond to each comment.

Reviewer 1

Comment R1.1: GCN Interpretability

"Lacks detailed analysis of the GCN component. Need investigations into correlations among multi-species sequences and enhanced interpretability of GCN-learned representations."

Response: We added a new Results section (Section 3.5, "GCN Interpretability and Species Importance") with an additive perturbation analysis: for each of the 58 extant placental mammal species, we replaced all other alignment rows with a gap token and measured the decrease in nucleotide cross-entropy compared to an all-gap baseline on 50,000 held-out variants (chromosomes 13 to 22). This quantifies the isolated information content of each species independently.

Key findings: (1) 53 of 58 extant species contribute positive isolated signal. (2) Human provides the largest contribution (ΔCE = 7.1). (3) Four diverged primates (Orangutan, Green monkey, Gorilla, Gibbon; ΔCE = 0.62 to 1.50) form a secondary cluster. (4) Chimpanzee ranks 11th (ΔCE = 0.062) due to information redundancy with the Human row (leave-one-out marginal $\Delta\text{CE} \approx -2.4 \times 10^{-6}$). (5) The SE-attention gate values are consistent with the perturbation rankings, showing that the model's internal weighting reflects phylogenetic structure. (6) Position-occlusion saliency confirms that predictions are anchored at the variant site, with sharper localization in the wider context model (Supplementary Figure S3).

New figures: Figure 6 (gcn_species_importance bar chart), Supplementary Figure S1 (tree ablation), Supplementary Figure S2 (model component ablation), Supplementary Figure S3 (position-occlusion saliency comparison).

Manuscript change: Methods Section 2.2 updated to describe center extraction (not mean pooling). Results Section 3.5 added. Discussion updated with interpretability summary.

Comment R1.2: Sequence Fragment Length (65 bp) Justification

"Fixed 65 bp not justified. Need systematic experiments to demonstrate optimality or robustness of this choice."

Response: We trained models with three context-window settings: flank $\in \{16, 32, 100\}$ bp (input sequences of 33, 65, and 201 bp). All models used the same training procedure on chromosomes 1 to 10 with chromosomes 11 to 12 as validation. The results (Table 2, Section 3.4) show that flank = 32 (65 bp) has the highest overall held-out AUROC (0.625) among the three settings, compared to the narrower flank = 16 (0.622) and the wider flank = 100 (0.617).

Manuscript change: Section 3.4 "Context Window Ablation" added with Table 2 reporting region-resolved AUROC for all three window sizes.

Comment R1.3: Complementarity Evidence

"Claim that GraphyloVar finds unique information not supported quantitatively. Need additional analyses to demonstrate complementarity and unique contributions."

Response: We added quantitative evidence of complementarity in Section 3.4. We computed a z-normalized score ensemble of GraphyloVar and CADD on the full held-out test set (chromosomes 13 to 22, $\sim$149M variants). The ensemble shows AUROC = 0.6442, an increase of +0.020 over GraphyloVar alone ($p < 10^{-15}$, DeLong test on 500,000 held-out variants). Bootstrap 95% confidence intervals (B = 1,000) do not overlap between GraphyloVar alone and the ensemble. Figure 5 plots the per-model and ensemble AUROC with confidence intervals.

Manuscript change: Section 3.4 "Complementarity" expanded with DeLong test statistics, bootstrap confidence intervals, and Figure 5 (fig_ensemble_auroc).

Comment R1.4: Evo2 and GPN-Star Comparison

"Must include quantitative comparison with Evo2 and GPN-Star."

Response: GPN-Star has been added as a baseline throughout all quantitative results. Evo2 was tested but requires GPU compute capability 8.9 or higher (Ada or Hopper class); our hardware is NVIDIA Quadro RTX 6000 (compute capability 7.5), which is below this requirement. We state this explicitly in Methods Section 2.4 and note Evo2 as an important future comparison point. GPN-Star results are included in Figures 2A, 3, and Table 1.

Manuscript change: Methods Section 2.4 updated with GPN-Star and Evo2 paragraphs. Evo2 exclusion justified by hardware constraint.

Comment R1.5: Code and Data Release

"Repository missing complete implementation and datasets. Must release full source code and datasets with usage procedures and licenses."

Response: A new section "Data and Code Availability" has been added at the end of the manuscript. It provides: (1) the GitHub URL for source code, trained model weights (flank = 32 checkpoint), and preprocessing scripts; (2) the dbGaP accession (phs000964) for the TOPMed whole-genome sequencing data; (3) the UCSC URL for the 100-way vertebrate alignments; and (4) pointers to the original publications for each MPRA dataset.

Manuscript change: "Data and Code Availability" section added (after Discussion).

Comment R1.6: Dataset Inconsistencies

"Saturation mutagenesis (Kircher et al. 2019) mentioned but not shown. GWAScatalog for 3'UTRs in Figures 3 and 5 but not described. Fix consistency."

Response: Kircher et al. (2019) is now included in Methods Section 2.5 as one of the three datasets forming the genome-wide MPRA group (along with Abell et al. 2022 and McAfee et al. 2023), and its use is described in the text. The 3'UTR group is now described explicitly: Griesemer et al. (2021) used GWAScatalog annotations to identify disease-associated loci, and Schuster et al. (2023) characterized prostate cancer 3'UTR variants; both are described in Section 2.5.

Manuscript change: Methods Section 2.5 rewritten to fully describe all 13 MPRA datasets grouped by the three categories shown in Figure 3.

Comment R1.7: MAF Correlation Figures

"Results reported only as text. Add supplementary figures or tables."

Response: The Spearman correlation values are now reported in Table 1 (Section 3.2), a formal table rather than in-line text. The table includes GraphyloVar ($\rho = 0.164$) and all five baselines.

Manuscript change: Section 3.2 now includes Table 1 (Spearman correlations) as a formal table.

Comment R1.8: Figure 1 Architecture Issues

"Mean Pooling to Attention inconsistency with GCN inputs. Unexplained white text near Concatenate Left + Right. Clarify or fix."

Response: The Figure 1 caption has been corrected to accurately describe the architecture: the model applies center extraction (not mean pooling) after the transformer blocks to obtain a 64-dimensional representation per species; the SE-gate then produces per-species attention weights before the GCN. The caption now explains each component in the order it appears in the figure.

Manuscript change: Figure 1 caption updated in full (Methods, Figure caption after line 142).

Comment R1.9: Figure 2 Caption Error

""GraphyloVar Score (dark purple) and GraphyloVar Score (purple)" is redundant. Fix caption, legend colors, labels."

Response: Fixed. The caption now reads "GraphyloVar Allele Ratio Score (dark purple) and GraphyloVar Combined Score (purple)", which correctly distinguishes the two model outputs.

Manuscript change: Figure 2 caption corrected (line 275).

Comment R1.10: Naming: GPN-MSA vs GPNMSA

""GPN-MSA" vs "GPNMSA": standardize throughout."

Response: We changed it to "GPN-MSA" throughout. We verified this with `grep` and there are no remaining "GPNMSA" instances.

Manuscript change: Global find-replace applied. Zero instances of "GPNMSA" remain.

Reviewer 2

Comment R2.1: Held-out Data Definition / Leakage

"Unclear if hold-out is variant-level or region-level. State explicitly. Region-based holdout evaluation recommended."

Response: The holdout strategy is now explicitly stated in Section 3.1: chromosomes 1 to 10 were used for training, chromosomes 11 to 12 for validation, and chromosomes 13 to 22 for the held-out test set exclusively. This chromosome-level partition avoids positional leakage. The test set contains approximately 149 million SNVs from the TOPMed whole-genome sequencing cohort.

Manuscript change: Section 3.1 first paragraph updated with explicit chromosome-level holdout description.

Comment R2.2: Prior Work: PHACT and PHACTboost

"Discuss PHACT (Kuru et al. 2022) and PHACTboost (Kuru et al. 2024) for phylogeny-aware variant prediction context."

Response: A discussion paragraph comparing GraphyloVar to PHACT and PHACTboost has been added to the Discussion section. The key point is that PHACT and PHACTboost require protein-level features (amino acid conservation, protein structure, functional domain annotations) and are therefore restricted to missense variants in coding regions. GraphyloVar targets non-coding variants using only multi-species whole-genome alignments. The two approaches are complementary rather than competing, and combining them in an ensemble for coding variants is an interesting direction for future work.

Manuscript change: Discussion paragraph added via included file (phact_paragraph.tex).

Comment R2.3: dbNSFP Comparison Breadth

"36 algorithms available in dbNSFP: compare to more, or justify exclusions."

Response: We have added a justification paragraph to Methods Section 2.4 explaining the baseline selection strategy. We selected baselines to represent methodologically distinct families: conservation metrics (PhyloP, PhastCons), integrative ML frameworks (CADD), sequence deep learning (Enformer), and MSA-aware models (GPN-MSA, GPN-Star). Running all 36 dbNSFP tools was outside our scope; these families collectively cover the major approaches to variant effect prediction and avoid redundancy.

Manuscript change: Methods Section 2.4 updated with method-selection justification paragraph.

Comment R2.4: Loss Function Clarification

"Allele-frequency head outputs categorical distribution but BCE reported as loss. Clarify."

Response: Corrected. Methods Section 2.3 now reads: "For the allele frequency head, we use categorical cross-entropy loss, which is appropriate for predicting a probability distribution over the five allele states (A, C, G, T, gap). For the SNP probability head, we use binary cross-entropy loss, since polymorphism status is a binary label." The original text incorrectly stated binary cross-entropy for both heads.

Manuscript change: Methods Section 2.3, training paragraph, corrected.

We think these revisions address the reviewer comments. We thank the reviewers again for their feedback.